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RESEARCH ARTICLE

Soft Barometric Tactile Sensor Utilizing Iterative Pressure Reconstruction

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ABSTRACT Tactile sensors play a pivotal role in enhancing robotic gripping systems across various industries, allowing robots to interact with their environment with precision and adaptability. This study introduces a novel soft barometric tactile sensor designed to capture pressure data beneath the contact pad within an elastomer layer. By employing an iterative fitting process for a parameterized pressure distribution, the sensor eliminates the need for extensive calibration or complex training procedures. A novel metric was introduced to evaluate the accuracy of pressure reconstruction in tactile sensors. This was complemented by quantitative investigations. Moreover, the precision of multiple contact detection and the capabilities of force estimation were examined. Finally, the practical applications of the sensor were demonstrated through in-hand measurement of object orientation. The sensor exhibits promising results, accurately reconstructing pressure distributions for a wide range of indenters, reliably discerning two contacts when their distance exceeds 5 mm and estimating forces with a Mean Absolute Error (MAE) of 1.83 N. Additionally, the in-hand reorientation achieved a Mean Absolute Rotation Error (MARE) of 6.81°, demonstrating the sensor's potential for real-world applications.

INDEX TERMS Tactile sensors, grasping, robotics and automation.

I. INTRODUCTION

Robotic gripping systems play a pivotal role in various industries, from manufacturing to healthcare and beyond, where dexterity and precision are paramount. In this context, tactile sensors serve as a fundamental component, enabling robots to interact with their environment, perceive surface properties and execute delicate gripping tasks with enhanced sensitivity and adaptability. In our human experience, tactile feedback plays an essential role, enabling us to execute intricate tasks effortlessly, often without reliance on visual cues. By integrating these tactile sensors into robotic grippers, a new dimension is added to their capabilities, allowing for the execution of more delicate tasks and the implementation of intelligent control that previously was unattainable. This integration of tactile sensing enables robots to interact with their environment with a sense of touch, significantly enhancing their ability to execute complex and precise maneuvers.

Various types of tactile sensors are employed, each utilizing distinct sensing principles. Vision-based tactile sensors [1], [2], [3], [4], [5], magnetic tactile sensors [6], [7] and barometric tactile sensors [8], [9], [10], [11] are commonly used sensing techniques. Additionally, other sensing techniques, such as capacitive or piezoelectric tactile sensors are also being explored in numerous applications. These applications span a wide spectrum, encompassing tasks ranging from dense box packing [12] or peg-in-hole insertions [13] to active contour following [14] and executing intricate in-hand manipulations [15]. These sensors are also indispensable in the medical field in application such as minimal evasive surgery [16] and prosthetics [17].

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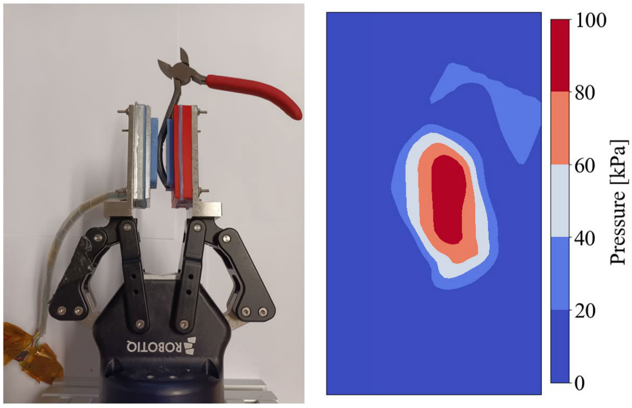


(a) (b)

FIGURE 1. **Tactile sensor mounted at the fingertips of a robotic gripper grasping pliers (a). Reconstructed pressure distribution (b).**

Additionally, there are numerous other applications in the biomedical field [18], as well as intelligent systems that incorporate tactile sensors [19].

The majority of existing studies in this field combine tactile sensors with various forms of machine learning. By utilizing barometric sensors, we directly measure a tactile feature, making interpretation easier and eliminating the necessity for extensive calibration or elaborate training procedures. This facilitates rapid and straightforward deployment in industry. Our work builds upon previous research [9], enhancing the sensor design significantly. This advancement enables the capture of finer details, removing the assumption of a point contact and allowing for the reconstruction of arbitrary shapes.

In this study, we introduce a soft barometric tactile sensor designed to capture pressure data beneath a contact pad within an elastomer layer. Figure 1 shows the sensor mounted at the fingertips of a robotic gripper. Employing an iterative process based on a parameterized pressure distribution enables the reconstruction of complex pressure distributions, accommodating both arbitrary shapes and multiple contact points. A novel metric was introduced to quantitatively examine the sensor's performance. Additionally, our investigation encompasses a qualitative analysis of pressure reconstruction accuracy, an assessment of multiple contact detection precision and an evaluation of force estimation. Furthermore, we demonstrate the capabilities of our approach through the in-hand measurement of an object's orientation.

II. RELATED WORK

A. TACTILE SENSORS

The topic of tactile sensors has been extensively studied over the last years. These tactile sensors use a variety of different sensing technologies. Recently, vision-based tactile sensors have gained a lot of popularity. These sensors infer high resolution tactile information, i.e. force and deformation, from camera images through the application of machine learning algorithms. One well-known example of such a vision-based tactile sensor is the Gelsight sensor [1]. In recent years, this design has been adapted for different applications, e.g., Gelsight360 [2] and GelSight Baby Fin Ray [3]. Another promising vision-based tactile sensor is the Insight sensor [4] which was recently converted into a miniature version, the Minsight sensor [5].

The second category comprises of magnetic tactile sensors. These derive tactile data by measuring changes in the magnetic field resulting from the deformation of a magnetized surface. Such sensors can perceive tactile information across varying levels of resolution, spanning from high [6] to low precision [7].

The final dominant sensing principle is barometric tactile sensing. This technique involves the direct measurement of pressures within elastomer layers utilizing Microelectromechanical Systems (MEMS) sensors. These sensors are generally used in contact localization [8], [9] and slip detection [9], [10], [11]. As a direct measurement of a tactile feature (i.e. pressure) is conducted, it allows for a more straightforward interpretation without the need for machine learning or extensive calibration procedures [9].

Additionally, other sensing principles (e.g., capacitive, piezoresistive and piezoelectric) can be used in tactile sensors. [20] provides an overview of tactile sensing utilized in robotic grasping.

B. TACTILE LOCALIZATION

Tactile sensors serve as valuable tools in localizing objects during grasping which enables functions such as slip detection and control, in-hand manipulation, etc.

In [21], a vision-based tactile sensor is employed to reconstruct the local patch map (i.e the deformation of the membrane of the tactile sensor) to recover the local geometry. The process involves training a model to learn a mapping from tactile images to generate a 3D reconstruction. This approach yielded a translational tracking error of 2 mm and a rotational tracking error of 10°.

Through the comparison of perturbed and unperturbed tactile images and the subsequent application of thresholding, [22] successfully determines the location and orientation of an object's contour during grasping. Additionally, they are capable of determining the normal force field and total shear forces. Their findings reveal a positional Root Mean Square Error (RMSE) of approximately 20 px and an orientation RMSE around 6°.

To localize an object during an in-hand rotation, [23] utilized a vision-based tactile sensor. Integrating the gradients observed in tactile images enables the reconstruction of a depth image. However, this process requires a calibration procedure.

C. SLIP DETECTION AND IN-HAND REPOSITIONING

Slip detection and control represent a significant application area for tactile sensors, primarily utilized to prevent object slippage [10]. Coupling these sensors with a supplementary

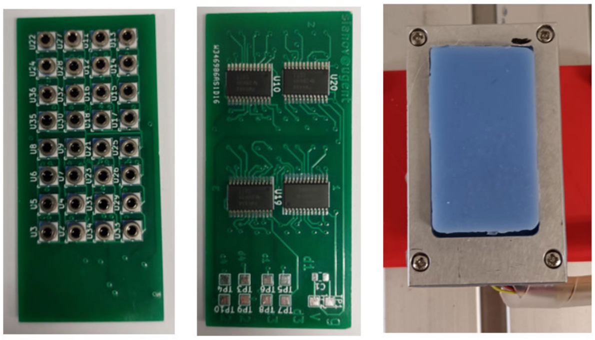


FIGURE 2. Sensor design. (a) PCB: top view. (b) PCB: bottom view. (c) Assembled tactile sensor.

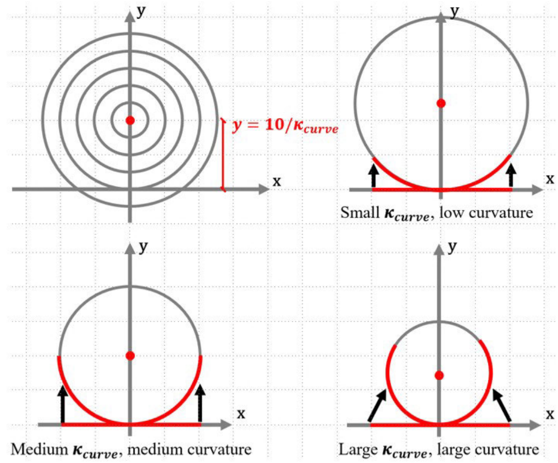


FIGURE 3. Curving of the rectangular pressure distribution.

vision system [24], [25] enhances their performance. An alternative approach involves intentionally accommodating slip during the grasp, enabling in-hand repositioning of an object.

In [26], an object is rotated in-hand solely using gravity. Through the utilization of an optical tactile sensor alongside a Long Short-Term Memory Neural Network, the rotation of an object to a specific target orientation was achieved. The analysis revealed a target error of 12.67°.

Using a Membrane Dynamics Model, [27] demonstrates the feasibility to drive a grasped object to a desired in-hand configuration solely using the environment. After this model was trained, a pivoting error of 5.41° was found.

In [28], a two-stage method was developed. First, the contact region is segmented from the tactile images using a Tactile Segmentation Neural Network. On this contact region an ellipse is fitted. By tracking the orientation of the primary axis of an ellipse a MARE of 1.85° was achieved. However slipping angles were limited to only 30°.

III. METHODS

A. SENSOR DESIGN

The design of our barometric tactile sensor takes inspiration from [9], but substantial improvements have been made. Specifically, we have increased the sensor count to a 4×8 array of digital micro altimeters (MS5840 [29]). Additionally, the spacing between sensors has been reduced to a center-to-center distance of 4.5 mm. These enhancements aim to capture finer pressure distribution details, enabling a transition from assuming single-point contact to accommodating arbitrary shapes with potentially multiple contact points. The sensors are soldered onto a 4-layer Printed Circuit Board (PCB) as depicted in Figure 2a. The bottom side of the PCB houses 4 multiplexers (TCA9548A) that read pressure data from the individual sensors (Figure 2b), substantially reducing the number of wires exiting the sensor.

Subsequently, an elastomer layer is prepared using a two-component liquid elastomer mixture, specifically Resion Resin Technology. Employing a vacuum pump, the elastomer is degassed and poured into a 3D-printed mold. In the final stage, the assembled tactile sensor, along with the elastomer layer in liquid form, undergoes degassing once more and is left to cure for 24 hours. This process solidifies the material into a silicon rubber with a shore index of 15A. Degassing is a fundamental technique in tactile sensor manufacturing, crucial for establishing a dependable bond between the protected MEMS transducer and the elastomer [30].

A final enhancement is the integration of a metal rim encircling the periphery of the contact pad (Figure 2c). This addition significantly increases the robustness of the bond between the pressure sensors and the elastomer, safeguarding against the detachment of the elastomer when the contact pad is subjected to tensile forces.

B. PRESSURE RECONSTRUCTION

Building upon our previous work [9], our aim was to devise a methodology capable of reconstructing various types of contacts, encompassing point, line, and surface contacts, utilizing a unified pressure distribution model. To accomplish this, we selected a rectangular pressure distribution, augmented with Gaussian drop-off to replicate the pressure dispersion inherent in the elastomer layer. This approach is mathematically represented as follows:

$$p(x, y) = \begin{cases} p_0, & \text{if } |x| < \frac{l_x}{2} \text{ and } |y| < \frac{l_y}{2} \\ p_0 e^{-\frac{1}{2}\frac{d^2}{\sigma^2}}, & \text{otherwise} \end{cases} \tag{1}$$

where $p(\cdot)$ is the pressure, p_0 is the pressure at the center of the rectangular distribution, l_x is the length of the primary axis of the rectangular pressure distribution, l_y is the length of the secondary axis of the rectangular pressure distribution, d is the minimal cartesian distance to the boundary of the rectangular pressure distribution and σ indicates the width of the Gaussian drop-off.

To reconstruct curved pressure distributions, we extend the existing pressure distribution model (1) with an additional parameter κ_{curve}. This extension involves curving the rectangular pressure distribution around a point on the y-axis with a coordinate $y = 10/\kappa_{curve}$ using polar coordinates. The factor 10 is introduced to enhance the parameter fitting. Figure 3 illustrates this process for the simplified scenario of a line contact along the x-axis which gets curved onto the circle with radius $10/\kappa_{curve}$. Employing this approach reveals that when $\kappa_{curve} = 0$, no curvature is introduced as the circle would have an infinite radius. However, as κ_{curve} increases, the distance to the center of the curvature decreases, resulting in a greater degree of curvature, as illustrated in Figure 3.

Ultimately, this pressure distribution can be positioned and oriented at any desired location on the contact pad through a final coordinate transformation as follows:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix} \begin{bmatrix} x - x_0 \\ y - y_0 \end{bmatrix} \tag{2}$$

where x' and y' are the transformed coordinates, α is the angle between the primary axis of the rectangular pressure distribution and Cartesian x-axis, x_0 is the Cartesian x-coordinate of the center of the rectangular pressure distribution and y_0 is the Cartesian y-coordinate of the center of the rectangular pressure distribution.

This culminates in a pressure distribution characterized as a curved rectangular shape with Gaussian drop-off. It is defined by eight parameters, namely:

- p_0 [Pa]: pressure at the center of the pressure distribution (i.e. the maximum pressure)
- σ [mm]: indicates the width of the Gaussian drop-off
- l_x [mm]: length of the primary axis of the rectangular pressure distribution
- l_y [mm]: length of the secondary axis of the rectangular pressure distribution
- κ_{curve} [mm^{-1}]: indicates the curvature
- α [°]: the angle between the primary axis of the rectangular pressure distribution and Cartesian x-axis.
- x_0 [mm]: the Cartesian x-coordinate of the center of the rectangular pressure distribution
- y_0 [mm]: the Cartesian y-coordinate of the center of the rectangular pressure distribution

To demonstrate the impact of these parameters on the pressure distribution, Fig. 4 presents artificially generated pressure distributions. All these parameters will from now on be represented by the single variable $\theta = [p_0, \sigma, l_x, l_y, \kappa_{curve}, \alpha, x_0, y_0]$.

The pressure reconstruction can be reformulated as an optimization task aimed at finding the optimal parameters of the pressure distribution that best aligns with the observed pressure data. This objective is achieved by using a least-squares minimization on the following error function:

$$Err(\theta) = \sum_{i=1}^{n} [p(x_i, y_i, \theta) - \widehat{p}_i]^2 \tag{3}$$

where $p(\cdot)$ is the estimated pressure, x_i and y_i are the x- and y-coordinate of sensor i, θ are the parameters of the pressure distribution and $\widehat{p}_i$ are the measured pressure values. Powell's dog leg method was selected as the optimization technique.

To enhance the performance of the optimization, the initial guess plays a crucial role. Given the assumption of limited changes between each time-step, the results from the previous time-step are used as the initial guess for the current time-step. If there was no contact during the previous time-step, a rough estimation of the parameters is made based on the raw pressure data.

After the optimal parameters θ^* are found, the residual pressures at each sensor location can be calculated as follows:

$$p_{res,i} = p(x_i, y_i, \theta^*) - \widehat{p}_i \tag{4}$$

where $p_{res,i}$ is the residual pressure of sensor i, $p(\cdot)$ is the estimated pressure of sensor i and $\widehat{p}_i$ is the measured pressure of sensor i.

The residual pressure values serve as indicators of the alignment between the current pressure distribution and the actual pressure readings. Elevated residual pressures at particular sensors indicate the presence of pressure distributions not accurately represented by the existing parameters θ^*. In the reconstruction of complex or multi-contact geometries, such elevated residuals are expected, as a single parameter set may not suffice to capture the intricacies of these geometries. To address this challenge, we employ an iterative approach where the parameter fitting process (3) is repeated using the residual pressures (4) instead of the measured pressure values. This iterative methodology is essential to capture these unaccounted pressure distributions and facilitate the reconstruction of arbitrary shapes. This transforms θ^* from a single parameter set into a list of multiple parameter sets. The final pressure distribution is obtained by summing all the individual pressure distributions, each characterized by their respective parameters listed in θ^*.

This iterative process continues until the desired accuracy is achieved or the maximum number of iterations is reached. Setting a maximum iteration limit becomes essential to enable real-time operation at a predefined frequency due to increased computation time per iteration. In our current setup, the maximum iteration count is set to 5 to ensure the system operates at 20 Hz.

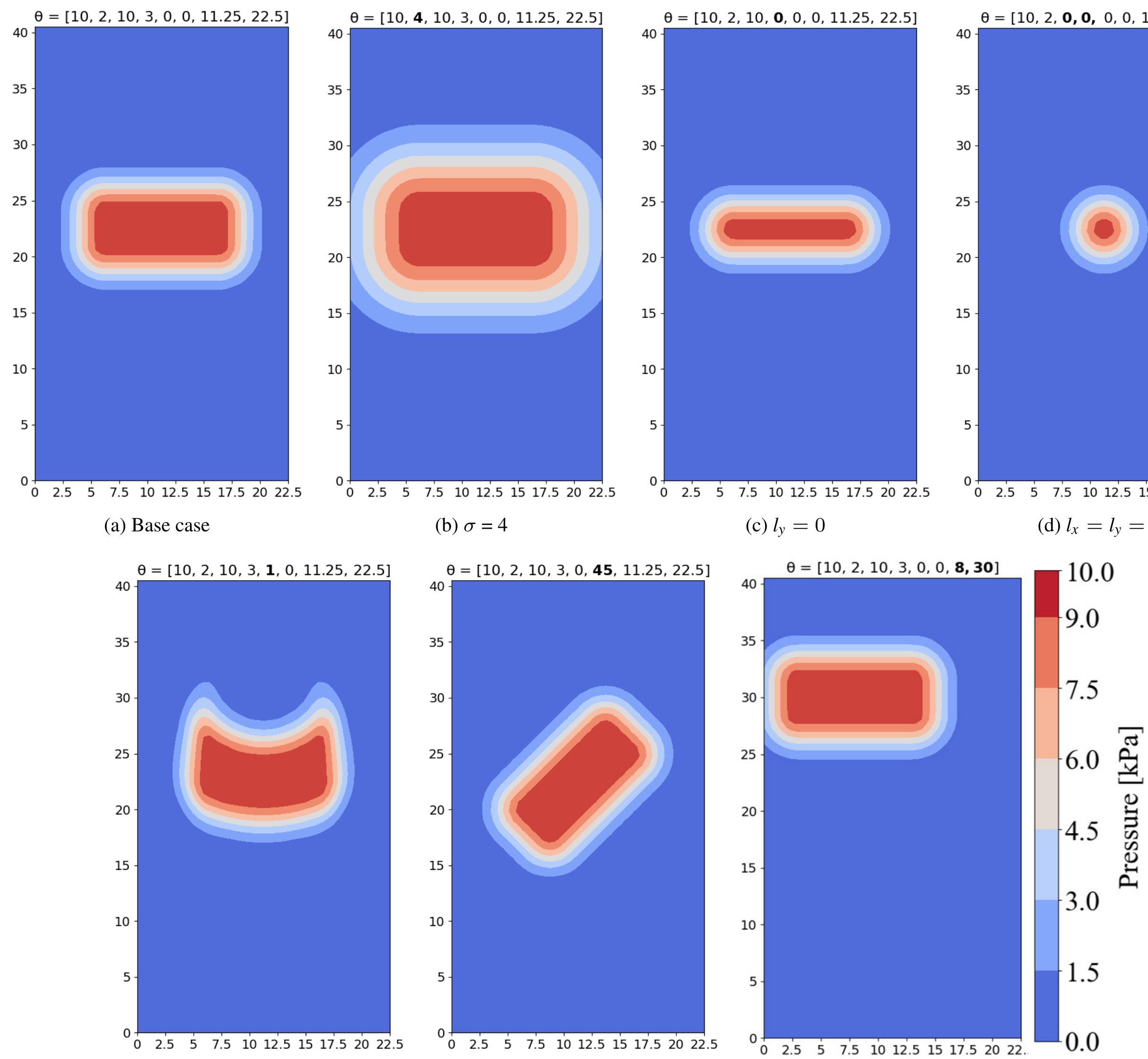
It is important to note that using a negative p_0 is permitted. This feature enables not only the addition but also the subtraction of pressure distributions from each other.

By integrating the final pressure distribution over the contact pad, the normal forces during contact can be found. In this case the integration is done numerically.

$$F_z = \int\int p(x, y, \theta^*)\, \mathrm{d}x\, \mathrm{d}y \tag{5}$$

IV. PRESSURE RECONSTRUCTION FIT

Assessing the accuracy of the pressure reconstruction (see Sec. V-B) poses a fundamental challenge in tactile sensor research. The true continuous pressure distribution beneath a contact pad remains inherently unknown, only measurable at discrete sensor locations. This prevalent issue lacks a universally applicable solution, prompting various strategies within the literature.



(a) Base case (b) $\sigma = 4$ (c) $l_y = 0$ (d) $l_x = l_y = 0$ (e) $\kappa_{curve} = 1$ (f) $\alpha = 45$ (g) $x_0 = 8$; $y_0 = 30$

FIGURE 4. Parameterised pressure distribution.

One such strategy involves comparing the sensor's total estimated force with the total measured force, providing a global quantitative measure of pressure reconstruction accuracy [22]. However, this approach fails to capture the local intricacies of the pressure distribution beneath the contact pad. Nonetheless Sec. V-D undertakes such a detailed examination with the aim to evaluate the accuracy of the force estimation (5).

Alternatively, a qualitative analysis of pressure reconstruction offers insights into accuracy by assessing multiple criteria and examining local details [2], [31]. However this approach does not allow for objective comparisons among different methodologies.

A third avenue involves constructing a Finite Element Method (FEM) model of the sensor [32], [33]. However this is beyond the intended scope of this study and may encounter similar difficulties in aligning the model with experimental data.

In this study, we propose a novel metric called the Pressure Reconstruction Fit (PRF) to evaluate the accuracy of pressure reconstruction. The PRF-algorithm, outlined in Algorithm 1, discretizes the domain in both Cartesian x- and y-directions (Lines 3-4) and iterates over the entire domain (Line 6). The PRF operates under the assumption that beneath a contact area, there exists a high-pressure region (Lines 7-11), whereas in regions without contact, pressure levels are low (Lines 12-17). This assumption guides the algorithm to increment the fit-value by 1 if the pressure reconstruction aligns with the expected contact conditions, and to maintain the fit-value if the assumption is not met. The PRF is then obtained by averaging this fit-value over all considered domain points (Line 19). This approach enables quantification of how

Algorithm 1 PRF-Algorithm

Require: $p(x, y, \theta)$; $(x, y)_{contact}$; p_{thres}; n

$\triangleright$ *Initialisation:*
$fit = 0$
$x_{sample} = \text{linspace}(0, x_{max}, n)$
$y_{sample} = \text{linspace}(0, y_{max}, n)$
$\triangleright$ *Calculation*:
for (x_i, y_i) in (x_{sample}, y_{sample}) **do**
  if (x_i, y_i) in $(x, y)_{contact}$ **then**
    $\triangleright$ Contact Region
    if $p(x_i, y_i, \theta) > p_{thres}$ **then**
      fit = fit + 1
    end if
  else
    $\triangleright$ No-Contact Region
    if $p(x_i, y_i, \theta) < p_{thres}$ **then**
      fit = fit + 1
    end if
  end if
end for
$PRF = fit/n^2$
return PRF

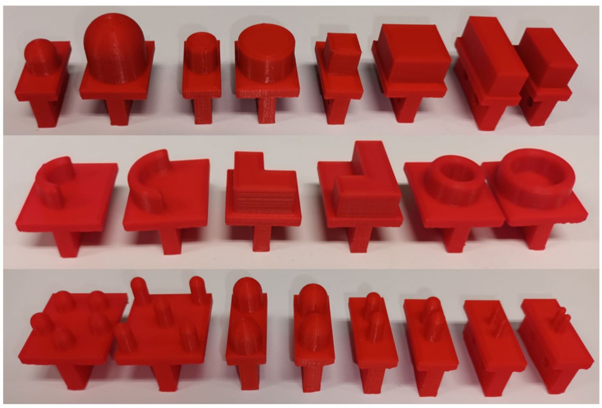


FIGURE 5. Indenters. Top row: Simple geometries. Middle row: Complex geometries. Bottom row: Multiple contact geometries.

effectively the pressure reconstruction captures the shape of the contact area.

V. RESULTS & DISCUSSION

A. DATA COLLECTION AND PROCESSING

In order to gather data for assessing the performance of the tactile sensor, a diverse set of indenters was 3D-printed. Firstly, there are the simple geometries encompassing spherical, cylindrical, square-shaped and rectangular indenters, available in both small and larger size. These are depicted in the top row of Figure 5. Secondly, the more complex geometries include C-shaped, L-shaped and O-shaped indenters, also available in small and larger size. These are displayed in the middle row of Figure 5. Finally, there are the multiple contact indenters, comprising two five-point contact indenters structured as a pentagon and an X-shape, alongside two-point contact indenters with varying distances between the contact centers, ranging from 16 mm to 4 mm. These are showcased in the bottom row of Figure 5. Some of these indenters can be seen in more detail in Figure 6.

TABLE 1. The average PRF for the different geometry types and different maximal iteration count.

	$it_{max} = 1$	$it_{max} = 2$	$it_{max} = 5$
Simple geometries	92.6%	92.7%	92.9%
Complex geometries	80.8%	82.1%	83.8%
Two-point contacts	86.8%	89.5%	89.7%
Five-point contacts	71.6%	75.4%	79.3%

These indenters are mounted at the end-effector of a position-controlled robotic arm (UR5e). The UR5e was controlled using the MoveIt! motion control package in a ROS environment (ROS Noetic patched with a Preemptive Linux kernel). This setup enabled continuous monitoring and recording of the position (x, y) and orientation (α) of the indenter throughout the experiments. Furthermore, a 6-Degree of Freedom (6-DoF) ATI M80-M8 force/torque sensor was employed to capture forces and torques in all three spatial dimensions. We kept record of the normal forces F_z (resolution of 0.04 N).

The tactile sensor produces 32 individual scalar pressure values at a rate of 20 Hz, which are organized into a vector $\widehat{\mathbf{p}}$. Figure 6 illustrates $\widehat{\mathbf{p}}$ for various indenters. It is important to emphasize that the pressure values $\widehat{\mathbf{p}}$ obtained from the barometric sensors must be reset to zero after each contact event. This compensates for (shear) stress introduced during repeated contact events at the bottom surface. Another essential implementation detail involves downsampling the position and force sensor readings to 20 Hz, as these are originally sampled at higher frequencies (up to 500 Hz) than the pressure data. Finally, it is important to note that the data was collected dynamically, without allowing the rubber to first adjust or settle to a new indenter. This decision was made to prioritize a fast response time, as we believe this is crucial for integrating the sensor with robotic applications.

B. EVALUATION OF THE PRESSURE RECONSTRUCTION

To assess the effectiveness of the pressure reconstruction algorithm, we compute the PRF for all indenters (Figure 5). For each indenter, four experiments were conducted wherein it was randomly positioned and oriented, with the force gradually increasing from 1 N to 20 N. The computation of the PRF requires four inputs: (i) The reconstructed pressure $p(x, y, \theta^*)$, as discussed in Sec. III-B. (ii) The contact location $(x, y)_{contact}$, derived from the position of the robot arm and the known shape of the indenter. (iii) The threshold pressure, defined as $p_{thres} = 0.333p_{max}$. (iv) $n = 50$. The results are presented in Table 1.

These results suggest that our method excels in accurately reconstructing the pressure field when dealing with simple geometries. In such cases, employing an iterative approach offers little advantage, as a single parameterized

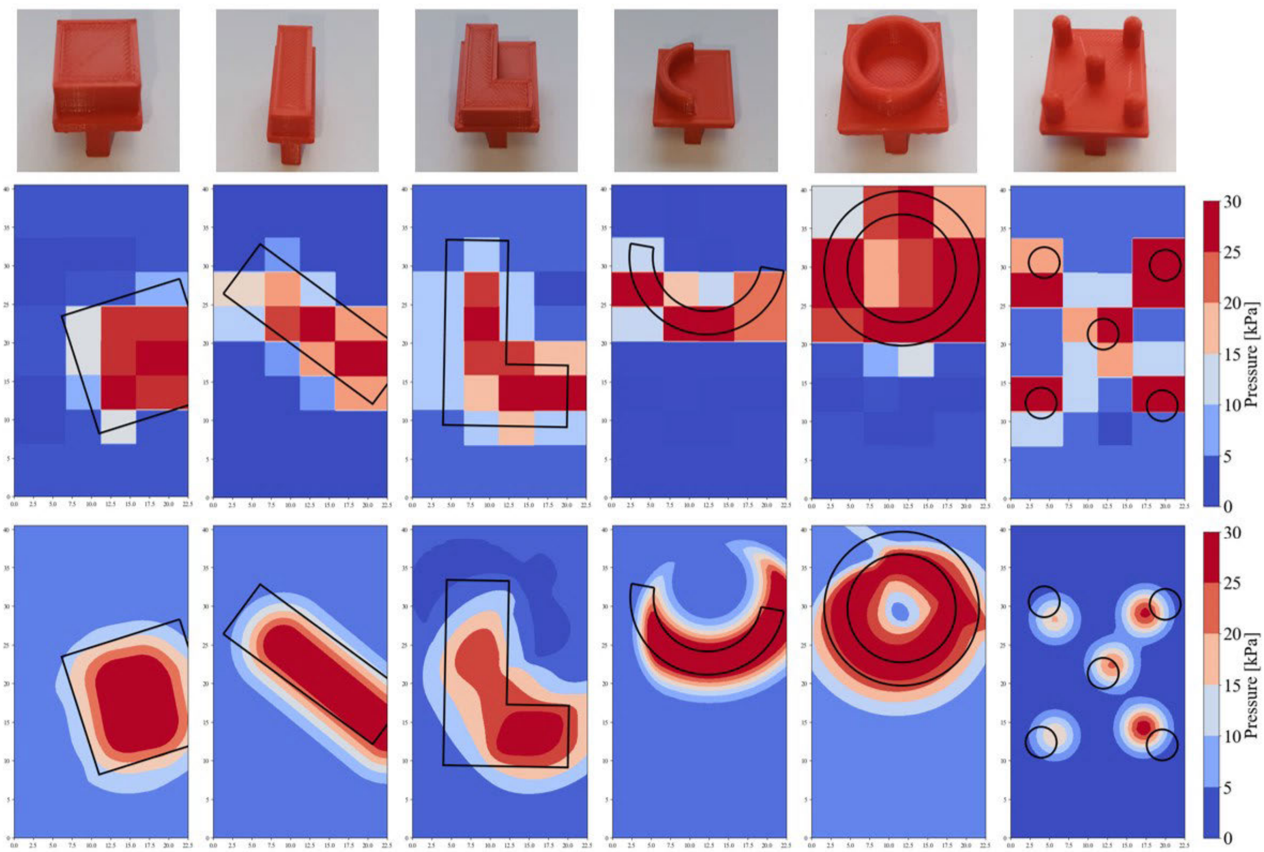


FIGURE 6. Raw pressure data (middle row) and reconstructed pressure distribution (bottom row) of different indenter contours (black). From left to right: square-shaped indenter, rectangular indenter, L-shaped indenter, C-shaped indenter, O-shaped indenter and five-point X-shaped indenter.

pressure distribution suffices to accurately represent the pressure field. Conversely, when tackling complex geometries, we observe a decrease in the average PRF. This outcome is expected given the inherent challenges associated with reconstructing such geometries. However, noteworthy is the observed increase in PRF with a higher maximum number of iterations. Though this improvement is somewhat limited, it underscores the potential effectiveness of our methodology.

When analysing the results of the multiple contact indenters, the influence of the maximum iteration count aligns with expectations. The two-point contacts demonstrate improvement with an increased iteration count to 2. Increasing the maximal iteration count even further has minimal impact on the PRF, as both contacts are already accounted for in the reconstructed pressure field. The PRF of the five-point contacts stands to benefit significantly from raising the maximum iteration count to 5. Here, with 5 contacts requiring reconstruction, the increased iterations prove advantageous, as anticipated. A more elaborate analysis of the multiple contact detection is conducted in Sec. V-C.

These results depend on the value of p_{thres}, as defined earlier in this section. Further analysis indicates that this influence is minimal when p_{thres} is maintained between $0.1p_{max}$ and $0.5p_{max}$. Although p_{thres}, primarily impacts the absolute PRF values, the relative differences between the various cases remain consistent. Therefore, the same conclusions can be drawn.

Supplementary to the quantitative analysis, a qualitative analysis was conducted. This involved comparing the reconstructed pressure distribution with the contour of the indenter's shape when poking the sensor with a randomly positioned and oriented indenter. Some of these results are shown in Figure 6. This figure distinctly illustrates that when dealing with simple geometries, such as square-shaped and rectangular indenters, the pressure reconstruction effectively captures the contours of the indenter, particularly regarding the object's position and orientation. Upon examining complex geometries, it is evident that not all intricate features, like sharp corners, are captured with equivalent precision. Moreover, the pressure reconstruction experiences a decline in precision near the edges of the sensor due to the reduced availability of pressure information in those areas. Nevertheless, the overall outline of each object remains distinctly recognizable. As mentioned earlier, the maximum iteration count was set at 5. This enables the detection of up to 5 distinct contact events, as depicted on the far-right side of Figure 6.

TABLE 2. Success-rate and average estimated distance between multiple contacts for multi-point contact indenters with varying distances.

Distance contacts	Success-rate	Average estimated distance
16 mm	100 %	14.1 mm
12 mm	99.9 %	9.7 mm
8 mm	100 %	5.6 mm
6 mm	84.5 %	5.5 mm
5 mm	61.7 %	4.6 mm
4 mm	28.3 %	4.2 mm

In Figure 6, a subtle observation reveals a slight bias in the sensor, particularly evident in the higher pressure readings on the right side compared to the left. This discrepancy is notably pronounced in the L-shaped and five-point contact indenters. The likely cause appears to stem from a slight variation in rubber thickness caused by an unlevel surface during curing of the elastomer, resulting in a marginally thicker rubber section on the right side.

The video attachment further demonstrates the performance of the pressure reconstruction.

C. MULTIPLE CONTACT DETECTION

The far-right section of Figure 6 clearly displays the sensor's capability to discern distinct contact events. This section aims to investigate the accuracy of detecting multiple contacts by employing the two-point contact indenters (Figure 5 - bottom row). Each indenter is subjected to four random positions and orientations, applying a variable force ranging from 1 N to 20 N on the sensor at each time-step. This resulted in 1200 samples for each indenter.

Two key metrics are used to assess precision. Firstly, the success-rate of multiple contact detection gauges the sensor's ability to identify the correct number of contact points. Secondly, if the correct number of contact points is identified, the distance between the two pressure distributions is calculated. The average estimated distance is compared against the actual distance of the indenter's contact points. Table 2 shows the results of the multiple contact detection.

When the distance between contacts measures 8 mm or more, consistent observations emerge. The success-rate for detecting multiple contacts reaches (nearly) 100 %, accompanied by an estimated distance error of approximately half the spacing between individual barometric sensors (i.e. 2.25 mm). However when the distance between contacts decreases, success rates start to drop. At a contact distance of 6 mm, the success rate drops to 84.5 %, reducing further to 61.7 % at 5 mm and significantly diminishing to 28.3 % at 4 mm. It is apparent that when the distance between contacts falls below the individual barometric sensor spacing (i.e. 4.5 mm), reliable discrimination between two contacts becomes unfeasible.

D. FORCE ESTIMATION

Understanding the behavior of the barometric pressure sensors is crucial before assessing the force estimation

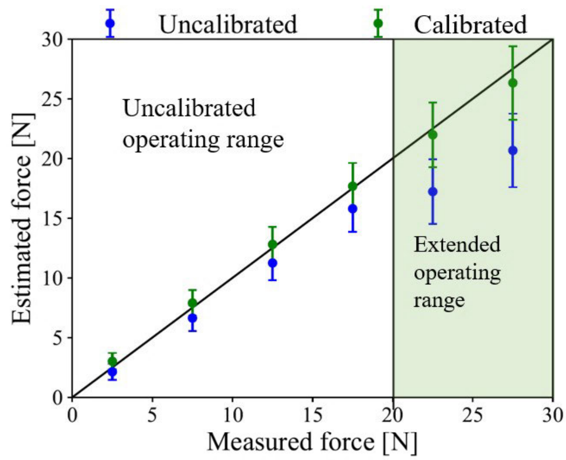
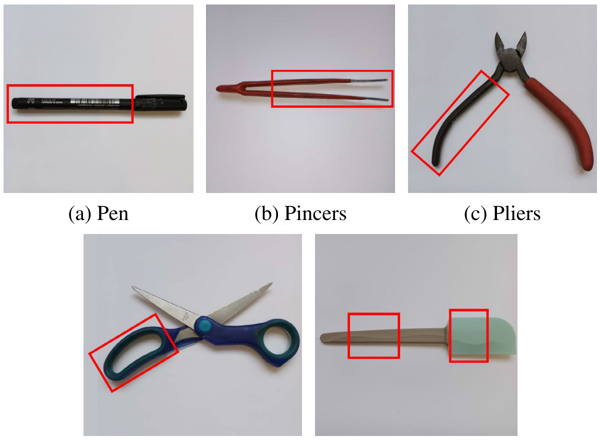


FIGURE 7. Measured versus estimated force.



(a) Pen (b) Pincers (c) Pliers

(d) Scissors (e) Spatula

FIGURE 8. Set of the 5 test objects.

accuracy. These sensors possess an extended pressure range, maxing out at 2000 mbar. Based on the performance characteristics, it can be inferred that when the sensor experiences pressures beyond its operating range, a negative pressure error is observed, which amplifies in magnitude as the pressure extends further beyond this range [29]. The resulting underestimation of pressure would consequently lead to an underestimation in force estimation and a decline in pressure reconstruction accuracy.

It is pertinent to note that the force at which this accuracy decline occurs varies depending on the indenter utilized. Smaller indenters concentrate force over a smaller surface area, resulting in higher pressures. These smaller indenters will therefore exhibit a narrower range of accurate force estimation behavior.

All the employed indenters underwent analysis, leading to the decision to cap the applied force at 20 N. In the case of smaller indenters, this force threshold often exceeded the pressure constraint. However, despite surpassing this threshold, the resulting impact on accuracy remained limited.

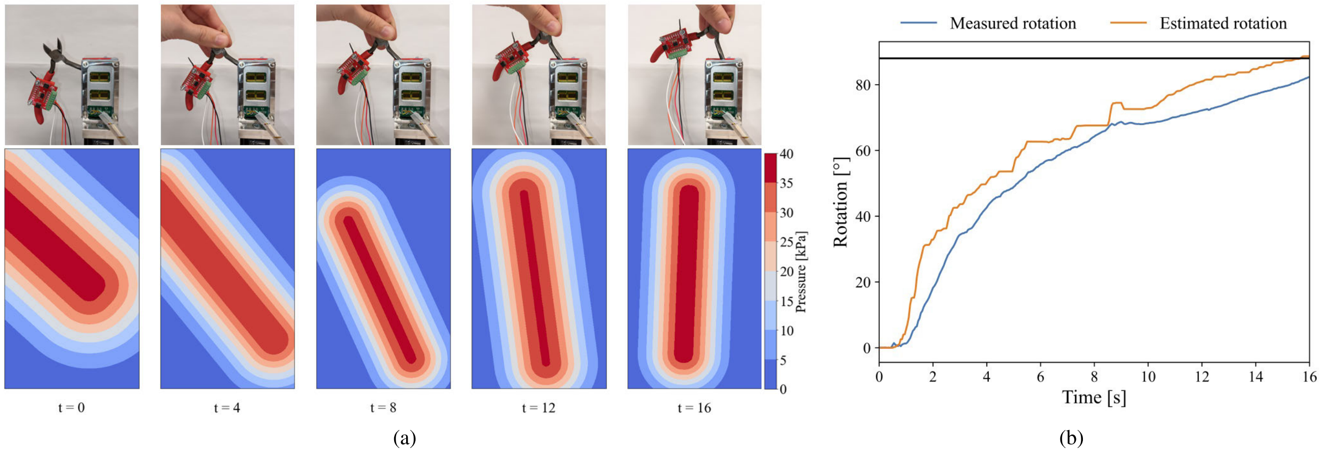


FIGURE 9. In-hand reorientation of pliers over 88°. (a) Experimental procedure (top). Pressure reconstruction (bottom). (b) Measured vs estimation rotation.

Figure 7 illustrates a comparison between the estimated force and the measured force across a slightly broader force spectrum. Notably, when forces remain below 10 N, the force estimation closely aligns with the measured data, yielding a MAE of 0.99 N. However, within the force range of 10 N to 20 N, it becomes evident that the estimated force tends to underestimate the actual force. This discrepancy can be attributed to some of the smaller indenters surpassing the pressure limit. This effect is only partially reflected on the MAE as it only applies to the smaller indenters. A MAE of 1.83 N is found. As the force increases to an even higher range (20 N to 30 N), this discrepancy becomes more pronounced, attributed to even the larger indenters exceeding the pressure limit. This is reflected in a higher MAE of 6.01 N.

For applications demanding greater precision in force estimation, a linear calibration procedure can be employed. This involves utilizing a 1-DoF force sensor with specific indenters and can be executed without the need for a robotic arm. As a result of this calibration, the MAE is reduced to 0.70 N, 1.02 N, and 2.86 N within the force ranges of 0 N to 10 N, 10 N to 20 N, and 20 N to 30 N, respectively. Consequently, the sensor's operational range is extended to 30 N, as indicated in Figure 7.

Even though the force estimation was limited to 20 N (or 30 N when calibrated), it is noteworthy that the sensor can withstand significantly higher accidental loads. In the experimental campaign, a load of 165 N was recorded without causing any damage to the tactile sensor.

VI. USE CASE: IN-HAND REORIENTATION

To showcase the capabilities of our sensor and the employed methodology, we establish a use case scenario wherein an object alters its orientation through an external disturbance while being held, eliminating the necessity for regrasping.

In this setup, an object is securely grasped by a robotic gripper (Robotiq 2F85) employing a predefined, object-specific gripping force. A PID-controller enables a constant gripping force through continuous actuation. Subsequently, a human intervenes and manually rotates the object. When the desired orientation is reached, the holding force is increased, immobilizing the object at its desired orientation. The change in orientation was assessed by comparing the α (Sec. III-B) of the current dominant pressure distribution with the α of the dominant pressure distribution before rotating. Here, the dominant pressure distribution is defined as the pressure distribution featuring the highest p_0 (Sec. III-B) if the iteration count exceeded 1. To quantify the orientation changes during each grasp, an Inertial Measurement Unit (IMU BNO080) is fixed to the object. Figure 8 illustrates the set of five test objects: a pen, pincers, pliers, scissors and a spatula. The red box indicates all possible grasp locations. Throughout the study, this experiment was repeated 9 times for each object, with a desired orientation change ranging from 10° to 90°.

TABLE 3. MARE of in-hand reorientation.

Object	MARE [°]	STD [°]
Pen	5.69	4.41
Pincers	5.28	5.71
Pliers	6.60	6.22
Scissors	5.86	7.54
Spatula	9.14	7.08
OVERALL	6.81	6.50

Figure 9 depicts one of these experiments, while a summary of all results can be found in Table 3. An overall MARE of 6.81° was found, displaying minimal discrepancies among the various objects. However, the spatula exhibited a slightly inferior performance. This discrepancy can likely be attributed to the elasticity of the rubber section (Figure 8e) - right square) causing slight movement at the handle, where the IMU was mounted.

Upon comparing these results to the current state-of-the-art, our approach showcases superior performance compared [26]. While our error aligns closely with [27], it is noteworthy that [28] outperformed our method. However, it is vital to highlight that [28] considered slipping angles only within the range of 0° to 30°, while our study covers slipping angles up to 90°. It is important to note that all these referenced works relied on machine learning, requiring an extensive training procedure. We successfully eliminate this step in our approach, saving considerable time and eliminating complexity.

VII. CONCLUSION & FUTURE WORK

The presented work introduces a tactile sensor capable of reconstructing pressure distributions beneath the contact pad. This parameterized reconstruction enables estimation of contact count, normal contact force and orientation changes during grasping. Direct pressure measurement beneath the contact pad enables fitting a parameterized pressure distribution without necessitating calibration or training procedures. For complex shapes, an iterative process is implemented to enhance accuracy.

Results demonstrate the reconstructed pressure distribution's alignment for a wide range of indenters. The sensor generally discerns two contacts when their distance measures 5 mm or more. Moreover, it demonstrates an ability to estimate forces with a MAE of 1.83 N without calibration, which improves to a MAE of 1.02 N with calibration. Furthermore, an in-hand reorientation controller exploiting this parameterized approach achieved a MARE of 6.81°.

Future developments aim to expand the model's capabilities by incorporating tangential force, indentation depth and torque measurements. Additionally, other pressure distributions besides the curved rectangular pressure distribution can be considered. Furthermore, incorporating translational slip into the existing orientation estimation poses a significant challenge because of the combined movements. Exploring advanced applications, particularly enhancing manufacturing tasks using tactile feedback, stands as a promising avenue for leveraging our sensor's capabilities in real-world scenarios. As tactile sensors prove versatile beyond gripping applications, exploring the implementation of the outlined methodology as a tactile skin around robotic arms or manipulators presents an intriguing avenue. This integration enables these systems to respond to and leverage contact with the environment to their advantage.

REFERENCES

[1] W. Yuan, R. Li, M. A. Srinivasan, and E. H. Adelson, "Measurement of shear and slip with a GelSight tactile sensor," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2015, pp. 304–311.

[2] M. H. Tippur and E. H. Adelson, "GelSight360: An omnidirectional camera-based tactile sensor for dexterous robotic manipulation," in *Proc. IEEE Int. Conf. Soft Robot. (RoboSoft)*, Apr. 2023, pp. 1–8.

[3] S. Q. Liu, Y. Ma, and E. H. Adelson, "GelSight baby fin ray: A compact, compliant, flexible finger with high-resolution tactile sensing," in *Proc. IEEE Int. Conf. Soft Robot. (RoboSoft)*, Apr. 2023, pp. 1–8.

[4] H. Sun, K. J. Kuchenbecker, and G. Martius, "A soft thumb-sized vision-based sensor with accurate all-round force perception," *Nature Mach. Intell.*, vol. 4, no. 2, pp. 135–145, Feb. 2022.

[5] I. Andrussow, H. Sun, K. J. Kuchenbecker, and G. Martius, "Minsight: A fingertip-sized vision-based tactile sensor for robotic manipulation," *Adv. Intell. Syst.*, vol. 5, no. 8, Aug. 2023, Art. no. 2300042.

[6] Y. Yan, Z. Hu, Z. Yang, W. Yuan, C. Song, J. Pan, and Y. Shen, "Soft magnetic skin for super-resolution tactile sensing with force self-decoupling," *Sci. Robot.*, vol. 6, no. 51, p. 8801, Feb. 2021.

[7] C. Ledermann, S. Wirges, D. Oertel, M. Mende, and H. Woern, "Tactile sensor on a magnetic basis using novel 3D Hall sensor–first prototypes and results," in *Proc. IEEE 17th Int. Conf. Intell. Eng. Syst. (INES)*, Jun. 2013, pp. 55–60.

[8] P. Piacenza, S. Sherman, and M. Ciocarlie, "Data-driven super-resolution on a tactile dome," *IEEE Robot. Autom. Lett.*, vol. 3, no. 3, pp. 1434–1441, Jul. 2018.

[9] T. De Clercq, A. Sianov, and G. Crevecoeur, "A soft barometric tactile sensor to simultaneously localize contact and estimate normal force with validation to detect slip in a robotic gripper," *IEEE Robot. Autom. Lett.*, vol. 7, no. 4, pp. 11767–11774, Oct. 2022.

[10] A. Grover, P. Nadeau, C. Grebe, and J. Kelly, "Learning to detect slip with barometric tactile sensors and a temporal convolutional neural network," in *Proc. Int. Conf. Robot. Autom. (ICRA)*, May 2022, pp. 570–576.

[11] G. Kim and D. Hwang, "Experimental characterization on slip detectability of barometer-based tactile sensor," *IEEE Sensors J.*, vol. 23, no. 18, pp. 21803–21811, Sep. 2023.

[12] S. Dong and A. Rodriguez, "Tactile-based insertion for dense box-packing," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Nov. 2019, pp. 7953–7960.

[13] S. Kim and A. Rodriguez, "Active extrinsic contact sensing: Application to general peg-in-hole insertion," in *Proc. Int. Conf. Robot. Autom. (ICRA)*, May 2022, pp. 10241–10247.

[14] U. Martinez-Hernandez, G. Metta, T. J. Dodd, T. J. Prescott, L. Natale, and N. F. Lepora, "Active contour following to explore object shape with robot touch," in *Proc. World Haptics Conf. (WHC)*, Apr. 2013, pp. 341–346.

[15] H. van Hoof, T. Hermans, G. Neumann, and J. Peters, "Learning robot in-hand manipulation with tactile features," in *Proc. IEEE-RAS 15th Int. Conf. Humanoid Robots (Humanoids)*, Nov. 2015, pp. 121–127.

[16] N. Bandari, J. Dargahi, and M. Packirisamy, "Tactile sensors for minimally invasive surgery: A review of the state-of-the-art, applications, and perspectives," *IEEE Access*, vol. 8, pp. 7682–7708, 2020.

[17] P. Saccomandi, E. Schena, C. Oddo, L. Zollo, S. Silvestri, and E. Guglielmelli, "Microfabricated tactile sensors for biomedical applications: A review," *Biosensors*, vol. 4, no. 4, pp. 422–448, Nov. 2014.

[18] M. I. Tiwana, S. J. Redmond, and N. H. Lovell, "A review of tactile sensing technologies with applications in biomedical engineering," *Sens. Actuators A, Phys.*, vol. 179, pp. 17–31, Jun. 2012.

[19] Y. Liu, R. Bao, J. Tao, J. Li, M. Dong, and C. Pan, "Recent progress in tactile sensors and their applications in intelligent systems," *Sci. Bull.*, vol. 65, no. 1, pp. 70–88, Jan. 2020.

[20] Z. Kappassov, J.-A. Corrales, and V. Perdereau, "Tactile sensing in dexterous robot hands—Review," *Robot. Auto. Syst.*, vol. 74, pp. 195–220, Dec. 2015.

[21] P. Sodhi, M. Kaess, M. Mukadanr, and S. Anderson, "PatchGraph: In-hand tactile tracking with learned surface normals," in *Proc. Int. Conf. Robot. Autom. (ICRA)*, May 2022, pp. 2164–2170.

[22] W. Kim, W. D. Kim, J.-J. Kim, C.-H. Kim, and J. Kim, "UVtac: Switchable UV marker-based tactile sensing finger for effective force estimation and object localization," *IEEE Robot. Autom. Lett.*, vol. 7, no. 3, pp. 6036–6043, Jul. 2022.

[23] W. Zhang, C. Xia, X. Zhu, H. Liu, and B. Liang, "TacRot: A parallel-jaw gripper with rotatable tactile sensors for in-hand manipulation," in *Proc. IEEE Int. Conf. Syst., Man, Cybern. (SMC)*, Oct. 2022, pp. 423–429.

[24] J. Li, S. Dong, and E. Adelson, "Slip detection with combined tactile and visual information," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2018, pp. 7772–7777.

[25] G. Yan, A. Schmitz, T. P. Tomo, S. Somlor, S. Funabashi, and S. Sugano, "Detection of slip from vision and touch," in *Proc. Int. Conf. Robot. Autom. (ICRA)*, May 2022, pp. 3537–3543.

[26] J. Toskov, R. Newbury, M. Mukadam, D. Kulić, and A. Cosgun, "In-hand gravitational pivoting using tactile sensing," 2022, *arXiv:2210.05068*.

[27] M. Oller, M. P. I Lisbona, D. Berenson, and N. Fazeli, "Manipulation via membranes: High-resolution and highly deformable tactile sensing and control," in *Proc. Conf. Robot Learn.*, 2023, pp. 1850–1859.

[28] J. Castaño-Amorós and P. Gil, "Measuring object rotation via visuo-tactile segmentation of grasping region," *IEEE Robot. Autom. Lett.*, vol. 8, no. 8, pp. 4537–4544, Aug. 2023.
[29] *MS5840-02BA*. Accessed: Dec. 4, 2023. [Online]. Available: https://www.te.com/usa-en/product-20000980-00.html
[30] J. W. Guggenheim, L. P. Jentoft, Y. Tenzer, and R. D. Howe, "Robust and inexpensive six-axis force–torque sensors using MEMS barometers," *IEEE/ASME Trans. Mechatronics*, vol. 22, no. 2, pp. 838–844, Apr. 2017.
[31] E. Roberge, G. Fornes, and J.-P. Roberge, "StereoTac: A novel visuotactile sensor that combines tactile sensing with 3D vision," *IEEE Robot. Autom. Lett.*, vol. 8, no. 10, pp. 6291–6298, Oct. 2023.
[32] C. Sferrazza, A. Wahlsten, C. Trueeb, and R. D'Andrea, "Ground truth force distribution for learning-based tactile sensing: A finite element approach," *IEEE Access*, vol. 7, pp. 173438–173449, 2019.
[33] G. Liang, D. Mei, Y. Wang, and Z. Chen, "Modeling and analysis of a flexible capacitive tactile sensor array for normal force measurement," *IEEE Sensors J.*, vol. 14, no. 11, pp. 4095–4103, Nov. 2014.

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